Sensing and Control of Single Trapped Electrons above 1 K

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Electrons trapped on the surface of cryogenic substrates (liquid helium, solid neon, or hydrogen) are an emerging platform for quantum information processing made attractive by the inherent purity of the electron environment, the scalability of trapping devices, and the predicted long lifetime of electron spin states. Here we demonstrate the spatial control and detection of single electrons above the surface of liquid helium at temperatures above 1 K. A superconducting coplanar waveguide resonator is used to read out the charge state of an electron trap defined by gate electrodes beneath the helium surface. Dispersive frequency shifts are observed as the trap is loaded with electrons, from several tens down to single electrons. These frequency shifts are in good agreement with our theoretical model that treats each electron as a classical oscillator coupled to the cavity field. This sensitive charge readout scheme can aid efforts to develop large-scale quantum processors that require the high cooling powers available in cryostats operating above 1 K.

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I. INTRODUCTION

Electrons trapped on the surface of superfluid helium are promising candidates for realizing mobile charge or spin qubits with long coherence times [1–3]. Similar to two-dimensional electron gases in semiconductor heterostructures, electron surface states form at the interface between two media, liquid helium and vacuum. The lateral electron motion can be controlled by nanofabricated gate electrodes beneath the liquid helium layer, which is typically 0.1–1 μm thick, to create electron traps similar to semiconductor quantum dots [4]. As for spin qubits in silicon [5,6], integration with CMOS control substrates is expected to enable the rapid scaling of electron-on-helium qubit systems.

Indeed, CMOS-fabricated chips covered with a helium dielectric layer have already been used to demonstrate the trapping and efficient clocked transport of charge "packets" containing small numbers of electrons [7,8]. In this architecture, the defect-free and nonmagnetic nature of liquid helium creates a pristine environment that isolates qubits from the host CMOS substrate, which is typically a major source of noise for silicon spin qubits [6,9].

The scaling of solid-state quantum information processing technologies to large qubit numbers comes at the cost of the heat load generated by external control lines and thus poses a significant technological challenge [10–12]. This presents a particular problem for superconducting qubits [13] and other systems that leverage superconducting circuit quantum electrodynamics (cQED) [14], which typically operate in dilution refrigerators that offer only limited cooling power (~1 mW at 100 mK) [15]. Other technologies, such as spin qubits in silicon, have shown more promise for operation above 1 K [16,17]. This allows operation in pumped ^{4}He cryostats in which cooling powers can exceed 100 mW [18,19]. For electrons on helium, while the charges remain bound to the helium surface at these temperatures (due to the large ground-state binding energy, $\Delta \approx 8$ K [20]) and the spin coherence time is expected to be extremely long [2,21], the control and readout of single electrons in these temperature regimes remains underdeveloped.

To date, only a limited number of experiments have demonstrated the detection and control of single electrons on the surface of helium. These include the use of sensitive single-electron transistor (SET) electrometers [22,23] and superconducting coplanar waveguide (CPW) resonators [24] to monitor the charge state of electron traps. The use of aluminum-based superconducting SETs in these studies constrained operating temperatures to below a few hundred millikelvin. Additionally, the dc operating regime employed in SET measurements limited the measurement bandwidth. Experiments utilizing superconducting CPW

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resonators, in which the electron trap was engineered to couple the electron orbital state to the resonator microwave field, were limited to temperatures close to 10 mK. These temperatures are standard in cQED experiments where coherent control requires the suppression of thermal excitations comparable with the operating frequencies (2–10 GHz) [24]. Such CPW resonators may facilitate electron spin readout via the induced coupling of the spin and motional degrees of freedom [3]. Other proposed spin readout schemes involve more advanced spin-to-charge conversion schemes, for example, parity readout schemes based on Pauli spin blockade [25], which can be adapted from the field of semiconductor quantum dots. Therefore, the development of fast and reliable single-electron readout schemes that operate close to 1 K is essential to the development of scalable quantum technologies based on electrons on helium.

In this work, we demonstrate the detection of single electrons on helium at temperatures above 1 K using a CPW resonator with an integrated electron trap. The trap is connected to a separate electron reservoir from which electrons can be deterministically loaded. At these temperatures, the thermal energy exceeds the electron motional frequency by several times, and the ^{4}He vapor pressure is non-negligible. This creates an electron environment substantially different to that in previous single-electron measurements. Despite these conditions we resolve reproducible resonator frequency shifts as the electron trap is loaded with single electrons. Compared with the previous electron-on-helium cQED experiments, our single electron-resonator coupling is twice as strong and the charge detection circuit is completely decoupled from the larger reservoir of electrons. We compare the measured frequency shifts with a classical model of the electron-resonator coupling in which the motion of the trapped electrons modifies the polarization of the trap volume, thereby increasing the resonator capacitance and reducing its resonant frequency. This model, developed in combination with finite-element modeling (FEM) of the trapping potential, reproduces the experimental frequency shifts to good accuracy. Our results lay the groundwork for exploring single electron-on-helium spin properties at elevated temperatures and can aid efforts to develop large-scale quantum processors with electrons trapped at the surface of noble gas substrates [1,2,26], as well as electrons in radio-frequency traps [27] or in semiconductor quantum dots [28].

The paper is arranged as follows. In Sec. II we describe the device used in the experiment and give details of the measurement setup and charge readout scheme. In Sec. III we present experimental results demonstrating the controlled loading and unloading of the electron trap, as well as deterministic single-electron control, and compare the observed frequency shifts with those predicted by the model. In Sec. IV we discuss the impact of these results before summarizing our conclusions in Sec. V.

II. ELECTRON TRAP AND READOUT SCHEME

All measurements were conducted in a continuous-flow ^{4}He cryostat (ICEOxford DRYICE 1.0K) at a stable base temperature of 1.1 K. The device is composed of a set of niobium electrodes patterned on a high-resistivity silicon wafer section using multilayer electron-beam lithography and chlorine plasma etching, thus creating microchannel structures, as shown schematically in Fig. 1(a). As is common in such devices, an array of microchannels connected in parallel (in this case some 40 microchannels each 1 mm long, 700 nm deep) provide a large surface area for electron storage. This storage zone is connected to the electron trap by a single microchannel 200 μm long and

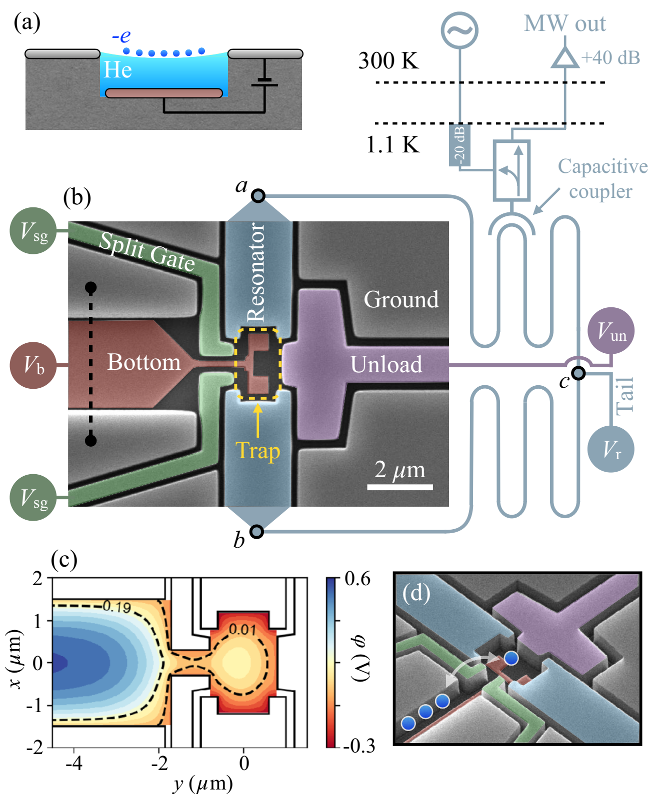

FIG. 1. (a) Schematic cross section [along the dashed line in (b)] of a microchannel filled with superfluid helium. The helium surface is charged with a system of electrons held on the surface by a potential difference applied between the upper and lower electrodes. (b) Schematic circuit diagram and false-color SEM image of the electron trap. For the differential resonance mode the ac voltage components at points a and b oscillate out of phase, with a voltage node established at point c. (c) Results of FEM calculation of the electrostatic potential $\varphi(x, y)$ in the plane of the helium surface. The dashed lines show isopotentials for 0.01 and 0.19 V, as labeled. The calculation is performed for typical trapping voltages: $V_b = 0.5$ V, $V_{sg} = -0.2$ V, $V_r = -0.3$ V, and $V_{un} = -0.5$ V. (d) Schematic depiction of the loading and unloading of the electron trap from the adjacent microchannel. The coupled electron–resonator system is driven by an incident microwave signal, and the reflected signal is measured (MW out).

3 μm wide, which we denote the electron reservoir. The microchannel network and the trap are filled by the capillary action of superfluid helium and the helium surface is then charged with electrons. Capacitive transport measurements are then used to confirm the presence of electrons in the storage zone and adjacent reservoir microchannel, and to estimate the electron surface density [29,30].

The electron trap is formed by the set of electrodes shown in Fig. 1(b) which are arranged to create a controllable potential well. In the plane of the helium surface the split gate electrodes (bias voltage V_{sg}) create a potential barrier between the trap and the electron reservoir, while the unload gate (bias voltage V_{un}) and the upper and lower arms of the resonator electrode (bias voltage V_{r}) provide electrostatic confinement. In the lower plane, at the bottom of the reservoir microchannel, the bottom electrode (bias voltage V_{b}) extends into the base of the trap region where it provides an attractive potential beneath the electrons. Further details of the device are given in Appendix A.

FEM is performed to calculate the electrostatic potential $\varphi(x, y)$ in the plane of the helium surface across the trap region [31,32] as well as the electron density n_{s} in the reservoir microchannel. As shown in Fig. 1(c), with suitable dc bias applied to each of these electrodes an electrostatic potential minimum for electrons is formed at the trap center, separated from the adjacent electron reservoir by a potential barrier. As demonstrated below, a small number of electrons can be loaded into the trap from the reservoir by varying the bias voltages on the trap electrodes, as depicted in Fig. 1(d). We note that the separation of the electron storage zone and reservoir from the readout resonator provides a clear advantage over previous devices in which electrons in both the trap and reservoir regions were coupled to the resonator [24,33].

As well as providing dc confinement, the resonator electrode forms part of the microwave circuit used to detect electrons in the trap. The CPW resonator (cavity) is designed to support a differential mode at bare resonance frequency $f_r = 6.04383$ GHz (see details in Appendix B). This particular resonator mode is described by out-of-phase ac voltage oscillations at the open ends [points a and b in Fig. 1(b)], where the electron trap is located. Voltage oscillations across the resonator electrodes produce an oscillating electric field that drives electron motion. The resulting collective electron motion within the trap alters the resonance properties of the cavity, which we detect by measuring shifts in the resonance frequency $\Delta f = f - f_r$ from the bare resonance frequency. In this setup, a probe microwave signal is introduced through a small capacitive coupling port into the cavity [see Fig. 1(b)], and the reflected signal is measured.

The influence of electrons in the trap on the resonator can be understood as follows: The collective oscillations of charges within the trap can be represented as polarized matter in the cavity described by electrical susceptibility $\chi_e(\omega)$. In the weak coupling limit (magnitude $\chi_e \ll 1$), this results in a shift in the resonance frequency of the cavity given by $\Delta f \approx -f_r \mathrm{Re}\chi_e(2\pi f_r)/2$. The electric susceptibility $\chi_e(\omega)$ characterizes the system's response to a perturbing microwave field and can be described as a sum of Lorentzian responses of discrete *vibrational* modes arising in the system of electrons. We find this quantity from the coupled equations of motion for electrons in the trapping potential $\varphi(x, y)$ under the external driving field (see details in Appendix C) and it takes the form

$$\chi_e(\omega) = \sum_n \frac{4g_n^2}{\omega_n^2 - \omega^2 + 2i\omega\Gamma_n}, \tag{1}$$

where g_n represents the coupling energy between the microwave field and the nth vibrational mode of the electron system, ω_n is the eigenfrequency of that mode, and Γ_n is the phenomenological damping constant. In order to calculate Δf for a given trapping potential and arbitrary number of electrons in the trap N_e, we first find the equilibrium configuration of electrons by energy minimization procedures. Subsequently, the frequencies and eigenvectors of the vibrational modes are obtained for a given electron configuration by numerically diagonalizing the total Hamiltonian of the electron system [34,35]. The coupling energy g_n is calculated based on the distribution of the electric field created by the resonator electrodes and the eigenvectors of the vibrational modes, completing the calculations of the resonator frequency shift. More details on the coupling between many electrons and the resonator appear in Appendix C.

III. EXPERIMENTAL RESULTS

A. Trap loading and unloading

Before investigating the electron-resonator coupling for a single electron we first demonstrate the controlled loading and unloading of a large number (several tens) of electrons into and out of the trap. As described in Sec. II, the electron trap is separated from the reservoir of electrons stored in the adjacent microchannel by a potential barrier formed between the split gate electrode. We perform a sequence of gate sweeps that systematically reduce the barrier height allowing electrons to spill into the trap, before raising the barrier again to corral a small number of charges within the trap (trap loading). The trap potential is then raised by sweeping the unload gate negative in order to push electrons back over the barrier until no electrons remain (trap unloading).

These loading and unloading procedures are depicted schematically in Fig. 2(a) with the aid of potential profiles generated by FEM. The corresponding experimental data are shown in Figs. 2(b) and 2(c) for trap loading and unloading, respectively. For the initial voltage configuration ($V_{\mathrm{b}} = 1.0$ V, $V_{\mathrm{sg}} = -0.4$ V, $V_{\mathrm{r}} = -0.3$ V, and $V_{\mathrm{un}} = -0.1$ V) a local potential minimum is formed in

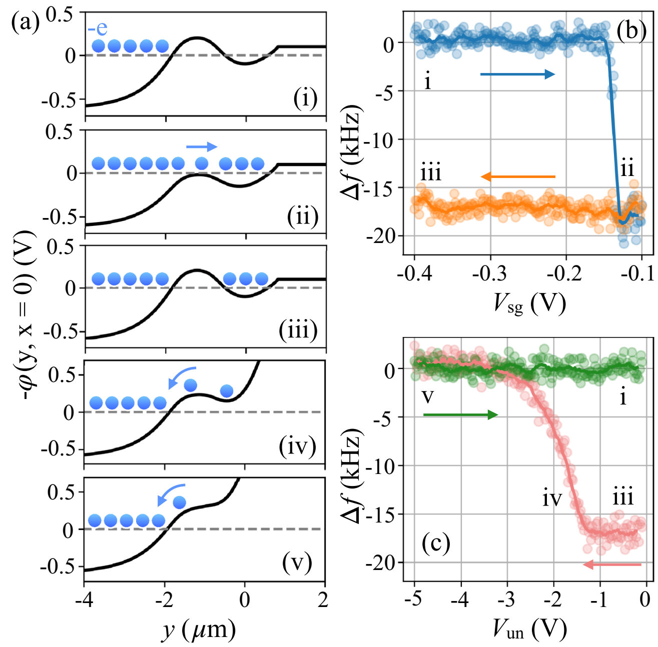

FIG. 2. Loading and unloading the electron trap at $T = 1.1$ K. For all plots $V_b = 1.0$ V and $V_r = -0.3$ V. (a) FEM analysis showing the evolution of the potential profile $-\varphi(x, 0)$ during the sequence of voltage sweeps used to load and unload the electron trap. To load the trap V_{sg} is swept from -0.4 V (i) to -0.1 V (ii) and back to -0.4 V (iii) with $V_{un} = -0.1$ V. To unload the trap V_{un} is swept to more negative voltage, with V_{sg} held at -0.4 V. For $V_{un} = -2.0$ V (iv) the trap potential is raised and some number of the trapped electrons escape to the adjacent reservoir, and for $V_{un} = -5.0$ V (v) the trap is emptied completely. Sweeping V_{un} back to -0.1 V returns the system to the initial condition (i). The measured Δf during loading and unloading is shown in (b) and (c), respectively. A ten-point rolling average is added as a solid line for clarity. Δf decreases when the trap is populated with electrons, in agreement with the electron-resonator coupling model.

the trap region, separated from the microchannel region by a potential barrier (i). Because the barrier height exceeds the value of the electron potential V_e, which is close to the ground potential 0 V, electrons cannot enter the trap. To allow electrons into the trap, V_{sg} is swept to the more positive value of -0.1 V. The height of the barrier is thereby reduced until it falls below V_e, allowing electrons to spill into the trap (ii). Because the trap area is negligible compared to that of the adjacent reservoir microchannel network, the electrochemical potential of the electron system remains unchanged and the electrons self-arrange within the trap geometry in order to maintain electrostatic equilibrium with the extended electron system in the microchannel. In agreement with the electron-resonator coupling model presented in Sec. II, f decreases as the trap is loaded with electrons. Once electrons enter the trap, V_{sg} is swept back to its initial value of -0.4 V. The potential barrier is restored, separating the trap from the microchannel, with some number of electrons remaining in the trap (iii). This completes the loading process.

Using the potential profile generated by FEM, numerical energy minimization procedures are employed to find the equilibrium electron positions within the trap [35]. For the voltage sweeps shown in Fig. 2, this analysis gives the total number of electrons loaded into the trap as $N_e \approx 30$. This number is reasonable given the typical surface electron density $n_s \sim 10^9$ cm^{-2} and the diameter of the trap potential well of approximately 1 μm. For these relatively large bias voltages the FEM analysis gives the trap curvature $\omega_e/2\pi \approx 50$ GHz, much higher than the bare resonator frequency. For these values of N_e and ω_e, our numerically calculated value of $\Delta f = -14$ kHz is in good agreement with the observed frequency shift of approximately $\Delta f = -17$ kHz.

Unloading is then performed by sweeping V_{un} negative [Fig. 2(c)]. As the electrochemical potential of the electrons in the trap rises to the level of the potential barrier, electrons escape to the microchannel and f increases (iv). For $V_{un} = -5.0$ V no potential minimum exists in the trap and all the electrons are expelled, and Δf returns to 0 kHz (v). Sweeping V_{un} back to its initial value of -0.1 V returns the system to the initial configuration with the trap empty (i). The loading and unloading procedures can then be repeated. We note that Δf decreases abruptly on electron loading because the relatively deep potential well is suddenly connected to the electron reservoir and quickly fills with electrons. In contrast, Δf changes smoothly during unloading because the depth of the trap and therefore the number of electrons within it decreases smoothly with V_{un}.

B. Deterministic electron control

Having demonstrated the controlled loading of the trap with several tens of electrons, we now reduce the trapping bias voltages in order to isolate a single electron. We also demonstrate that careful tuning of the confinement potential is required to bring the electron motion close to resonance with the cavity field in order to maximize the resulting Δf.

In Fig. 3(a) we show Δf as electrons are unloaded from the trap for $V_b = 0.3$ V. These measurements are performed in a similar manner to the loading and unloading curves shown in Fig. 3 with one important modification: After setting the unload gate to each increasingly negative voltage (which we denote V_{un}^{max}) the value is returned to the initial voltage of $V_{un} = -0.4$ V. Δf is then measured as V_r is swept between -0.09 and -0.13 V in order to vary the curvature of the trapping potential. Returning the unload gate bias to the same value while V_r is swept ensures that the resonator frequency is always measured under the same bias conditions, eliminating any dependence of the electron-resonator coupling on the position of the electron system in the x, y plane.

As in Fig. 2(c), sweeping V_{un}^{max} negative reduces the number of electrons in the trap. However, for this reduced

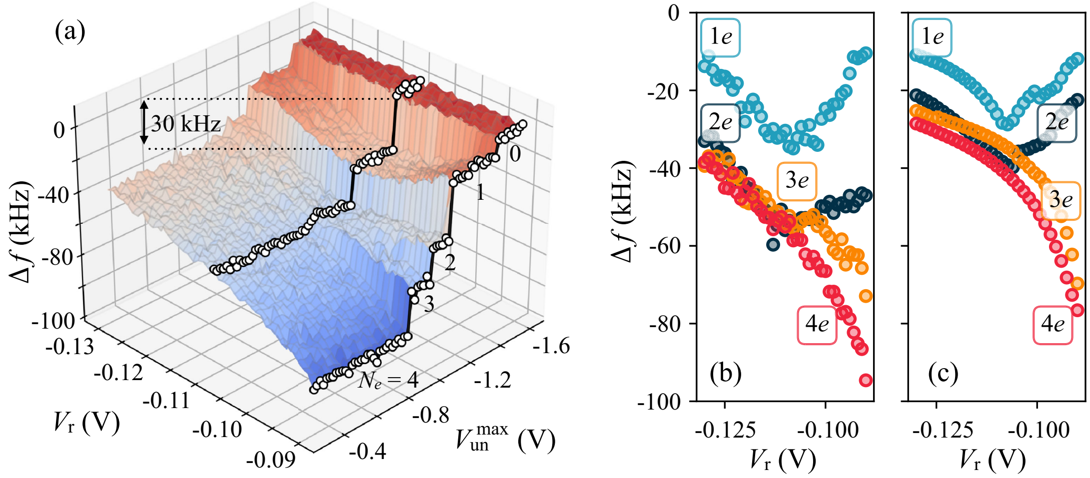

FIG. 3. (a) Surface plot of Δf with varying V_r and V_{un}^{max}. Two datasets corresponding to V_{un}^{max} sweeps at constant V_r are highlighted (white circles) for $V_r = -0.09$ V and $V_r = -0.11$ V, in order to show clearly plateaus in Δf corresponding to discrete N_e values and the maximal frequency shift for single-electron trapping. (b) Slices of Δf along V_r for the $N_e = 1, 2, 3, 4$ plateaus in (a). (c) Δf calculated using the expression for the electric susceptibility $\chi_e(\omega)$, for the same conditions as in (b).

bias configuration and smaller voltage step size, discrete and irreversible shifts in Δf are observed, which we attribute to changes in the number of trapped electrons. As a result, four plateaus in Δf are resolved, with the last plateau corresponding to a single isolated electron. However, the shape of each plateau depends strongly on the confinement curvature. For $N_e = 1$, the absolute value of Δf reaches a maximum of ~30 kHz for $V_r = -0.11$ V. As will be shown below, the maximum in $|\Delta f|$ corresponds to the case in which the trap curvature $\omega_e/2\pi$ is closest to the resonance frequency of the cavity.

The experimentally measured Δf exhibits two qualitatively distinct behaviors at the extreme values of V_r as the number of electrons in the trap is varied. For $V_r = -0.09$ V large discrete shifts in Δf are observed. In contrast, for $V_r = -0.13$ V discrete shifts are not clearly observed for $N_e > 2$. Our eigenmode analysis of the electron system indicates that the primary modes contributing to the electric susceptibility for $N_e = 2, 3, 4$ originate from the in-phase motion of all electrons (in-phase mode). For $N_e = 3$ (4) we find that the frequency of the in-phase mode approaches the resonator frequency ω_r at $V_r \approx -0.09$ (-0.08) V, resulting in the significant frequency shifts observed in the experiments as we vary the number of electrons in the dot. For $V_r = -0.13$ V, the mode frequencies are detuned significantly (13.6 GHz for $N_e = 3$ and 14.4 GHz for $N_e = 4$), thus reducing the system response. This results in only small changes in χ_e as N_e is varied. Our calculations of Δf based on Eq. (1) reproduce the experimental data with reasonable accuracy, as shown in Fig. 3(b) and 3(c).

Interestingly, the dependence of Δf on V_r for two electrons exhibits a similar response as the single-electron case. We measure values of Δf that are twice as large as those in the single-electron case for $V_r < -0.11$ V. This aligns with the predictions of our model which, for a harmonic approximation of the trap potential, indicates that the electron-resonator coupling energy for the in-phase motion of two electrons scales as $g_{2+} = \sqrt{2}g_e$, with a frequency equal to that for the single-electron case (see details in Appendix D). This approximation is more accurate for $V_r < -0.11$ V, where our FEM calculations predict a trap potential with a single minimum. For $V_r > -0.11$ V, however, the FEM results indicate that the single-well potential transforms into a double-well structure, introducing significant nonlinear terms into the potential. As a result, an exact factor of 2 is no longer expected, consistent with our experimental observations.

In Fig. 4(b) we demonstrate the repeated loading and unloading of the trap with one and two electrons. During the measurement, the bias values are chosen to give the maximal shift in $|\Delta f|$, resulting in repeated ~22 and 44 kHz shifts for one- and two-electron loading, respectively (the bias configurations employed are described in the caption). The observed frequency shifts agree well with calculated values of 20 (48) kHz for $N_e = 1$ (2), as predicted by our model. We observe no errors in this single- and double-electron control cycle. This measurement confirms reproducible single-electron control with good detection sensitivity. This is unsurprising given the helium surface is free of defects and that the energy required to escape the potential well (which is 60 mV deep according to FEM) greatly exceeds the electron temperature ($k_B T/e \approx 0.1$ mV). We note that these single-electron measurements are made without optimization of data acquisition speed. In addition, without amplification at the cryogenic stage our measurement is limited by noise from the room temperature amplifiers. Therefore, significant improvements in detection speed and sensitivity can be achieved in future experiments, which may be aided by

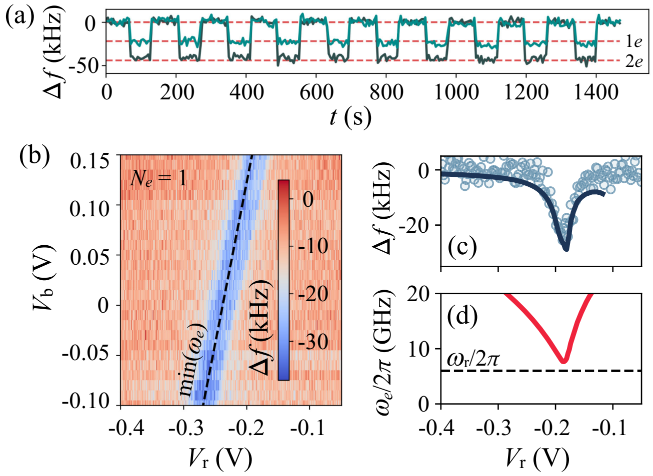

FIG. 4. (a) Δf recorded over time t during repeated loading and unloading of one and two electrons. All data points are recorded for the voltage configuration $V_\mathrm{b} = 0.3$ V, $V_\mathrm{r} = -0.1$ V, $V_\mathrm{un} = -0.4$ V, and $V_\mathrm{sg} = -0.5$ V. Electron loading and unloading is performed by applying a series of voltage step sequences on a timescale shorter than the measurement period (and therefore between data points). For electron loading $V_\mathrm{sg} = -0.4\ \mathrm{V} \rightarrow -0.05\ \mathrm{V} \rightarrow -0.4$ V with $V_\mathrm{r} = V_\mathrm{sg} = -0.4$ V and for unloading $V_\mathrm{un} = -0.4\ \mathrm{V} \rightarrow -3.5\ \mathrm{V} \rightarrow -0.1$ V with $V_\mathrm{r} = V_\mathrm{sg} = -0.4$ V. The horizontal dashed lines show frequency shifts of 22 and 44 kHz. (b) Δf against V_r and V_b for $N_\mathrm{e} = 1$. Here $V_\mathrm{un} = -0.4$ V and $V_\mathrm{sg} = -0.5$ V. The voltage configurations for which ω_e reaches its minimum are given by the dashed line, with a small +50 mV offset added to V_r to aid comparison with the experimental result. (c) Δf against V_r for $V_\mathrm{tr} = 0.15$ V. The solid lines are generated from Eq. (1) using potential profiles generated by FEM, as described in the text. (d) The trap curvature $\omega_e/2\pi$ for the same conditions as in (c).

more advanced measurement techniques such as homodyne sideband detection under gate voltage modulation of the electron signal [36].

The dependence of Δf on V_r occurs due to the variation of the confinement curvature, and according to Eq. (1) the maximum shift in Δf occurs when the single-electron frequency ω_e approaches closest to the resonator frequency ω_r. As the trap potential continuously evolves from a single-well to a double-well structure, we expect the trap curvature ω_e to decrease, reaching a minimum at certain values of V_r and V_b, and then increase. We confirm this in Fig. 4(c) by measuring Δf while varying V_r and V_b for $N_e = 1$. Because the curvature of the confinement potential depends on both voltages, the minimum in Δf moves along a linear path in the V_r-V_b plane. We also show the line of minimum ω_e, calculated using FEM, which closely follows the experimental dip. The good agreement confirms that ω_e is accurately controlled by tuning the confining bias voltages. In Fig. 4(c), we present the same frequency shifts as a function of V_r, measured at fixed V_b, alongside predictions from our model based on Eq. (1) (solid curve) with single-electron coupling strength $g_\mathrm{e}/2\pi = 9.8$ MHz, showing good agreement. The calculated values of ω_e for these voltage configurations are also shown in Fig. 4(d). We note that the value of g_e for our device is a factor of 2 larger than those previously reported in cQED experiments with electrons on helium [24].

The observation of a single minimum in the frequency shift suggests that the trap curvature remains higher than the resonator frequency ($\omega_e > \omega_r$) as the potential morphs from a single-well to a double-well structure. Given the evolution of the potential profile, we initially expect to observe two spectral crossings as the control voltages are varied: the first as the single well flattens and the electron motion passes through resonance with the resonator, and the second as the electron becomes localized in one of the double-well minima. A hallmark of these crossings would be a sign reversal in the frequency shift as the system passes through each resonance. However, these experimental signatures were not observed in any of our measurements and we consistently detected only a single dip in the resonator frequency as the electrostatic potential was varied. We therefore conclude that full resonance could not be reached and the electron motion was always probed in the dispersive regime. We find that this scenario is possible only if a small uncompensated electric field exists along the x axis (modeled in our study by applying a 20-mV difference between the two split gate electrodes). Such a field could arise from a slight misalignment of the upper and lower electrode layers during the fabrication process.

IV. DISCUSSION

We note that the electron system susceptibility given by Eq. (1) is valid only under the assumption of small fluctuations around the ground-state electron configuration. To validate this approximation, we estimated the average electron-electron distance and calculated the ratio of the average Coulomb energy to the thermal energy, $\Gamma_\mathrm{plasma} = U_C/k_\mathrm{B}T$, for the voltage parameters used in Fig. 3. We find that $\Gamma_\mathrm{plasma} > 30$ under these conditions, indicating that thermal energy is insufficient to disrupt the electron configuration, which thus remains in its ground state. Additionally, we performed numerical simulations of possible metastable configurations of the electron system and found no metastable states for small electron clusters with $N_e < 8$ under these conditions (see details in Appendix E).

The measured shift in the resonance frequency for a single electron can be understood as a renormalization of the resonator capacitance by an effective *single-electron capacitance*, analogous to the concept of quantum capacitance [37], defined as $\delta C_e = \chi_e(\omega)C \propto (\omega_e^2 - \omega_r^2)^{-1}$ in the dispersive regime, where C is the total capacitance of the resonator. For the largest frequency shift shown in Fig. 4(c), we estimate $\delta C_e \approx 0.45$ aF. To contextualize this value, we compare it to other measurement techniques. For instance, $\delta C_e \approx 10^{-3}$ aF was estimated for the out-of-plane motion of a single electron from many electron image-charge Rydberg state detection methods [38]. For single electrons in semiconductor quantum dots $\delta C_e \approx 1$–10 aF has been reported [39,40] using rf reflectometry. Our measured values of δC_e can be improved further by bringing the electron motion fully into resonance with the cavity field, bringing our measurement technique to the level of the most advanced rf SET charge sensors. Additionally, our theoretical model can be applied generally to LC resonator circuits capacitively coupled to bound electrons, which have been proposed for a variety of quantum computing architectures [26,27,41,42].

The ability to detect single electrons and electron clusters at elevated temperatures could open investigations of finite size effects on phase transitions from disordered, liquidlike states, to Wigner molecule formations [24,30]. For electron clusters, the characteristic plasma parameter, which quantifies the ratio of average Coulomb energy $\propto\sqrt{n_\mathrm{s}}$ to thermal energy $\propto T$ of electrons, can be adjusted by varying the temperature or the confining potential. The simplest scenario in which Coulomb interactions influence the properties of an electron system can be explored and tested is the case of two electrons. While in our experiments the in-phase motion of two electrons in the trap is coupled to the differential mode of the resonator, in general the out-of-phase motion of two electrons can be driven by the common mode. Therefore, the accurate engineering of the resonance frequency of both the common and differential modes of the resonator within the measurement frequency band could allow investigation of correlated electron motion driven by Coulomb interactions. Operating at low temperatures ($k_\mathrm{B}T \ll \hbar\omega_n$) and in the strong coupling regime ($g_n \geq \Gamma_n$) would make it possible to study entanglement in a two-electron system [43].

In our experiments, we did not observe statistically significant changes in the resonator linewidth that would allow direct measurement of the damping rates of the electron motion. However, we found that using $\Gamma_n/2\pi \approx$ 1–2 GHz provides more accurate agreement between the experiment and our model. Studies including electron mobility measurements [44] and Rydberg state spectroscopy [45] have shown helium gas atom scattering to be the dominant relaxation mechanism for 2D electrons on helium for $T \gtrsim 0.7$ K. A quantum-mechanical description of relaxation rates in this regime assumes that for free in-plane motion there are many available states to which an electron may scatter during interactions with helium atoms [20,46]. This condition will likely change for a single electron with strongly confined in-plane motion leading to a confinement-induced suppression of the relaxation rate. In future experiments, the use of resonators made from high-kinetic inductance materials, which increase electron-resonator coupling, along with improvements in the measurement readout scheme, could enable the extraction of single-electron relaxation rates and facilitate investigation of electron scattering mechanisms at temperatures above 1 K.

V. CONCLUSION

In summary, we have demonstrated the implementation at temperatures above 1 K of a CPW resonator as a charge sensing element for single electrons trapped on the surface of liquid helium. While the device and measurement circuit are familiar from cQED experiments, we have developed a classical description of the resonator-electron coupling for this elevated temperature that accurately predicts the observed frequency shifts for both large electron ensembles and single electrons. The shift in the resonance frequency of our readout resonator occurs due to the change in electric susceptibility that arises from the induced dipole moments of the vibrational collective modes within the electron ensemble. The level of electron control demonstrated in this work, together with engineering strong coupling between a many-body electron system and a cavity mode [34], could provide a new platform for exploring quantum optics models of light-matter interactions. This includes models such as the quantum Rabi, Dicke [47], and Hopfield [48] models, along with newer frameworks that extend beyond these traditional models [49,50].

Electrons on helium exhibit high mobility and extremely low spin-orbit interaction [2], making them a promising candidate for exploring *mobile* qubit architectures. Although replacing the liquid helium with a thin (∼10-nm) layer of solid neon has been shown to improve charge state coherence times [51,52], this is achieved at the cost of severely reduced trapping reproducibility due to the localization of electrons by random surface roughness [53,54]. A significant aspect of our work is the demonstration of the ability to reliably load and unload single electrons from the reservoir into the resonator measurement circuit—an essential feature of mobile qubit architectures. Further work toward fast electron operations and detection methods with improved bandwidth and signal-to-noise characteristics, and with smaller resonator footprints, will improve upscaling of electrons-on-helium-based quantum devices.

ACKNOWLEDGMENTS

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APPENDIX A: DEVICE AND EXPERIMENT

The device is fabricated by patterning a set of electrodes (niobium, thickness 80 nm), including the resonator electrode, on to the substrate before the silicon is selectively etched to create a system of trenches some $h = 700$ nm deep. A second set of electrodes (niobium, 30 nm) are then patterned at the bottom of these trenches. In the final fabrication step aluminum air bridges [55], each several tens of micrometers in length and several micrometers in height, positioned far (at least 100 μm) from the trench network are used to make necessary electrode interconnects.

As shown schematically in Fig. 1(a), this process allows the fabrication of microchannel structures in which the electrostatic potential is controlled by the integrated electrodes. In our experimental cell, which contains a small reservoir containing 0.2 cm^3 of liquid helium ~2 mm beneath the device, a thin (~30-nm) film of superfluid helium covers all internal surfaces including the upper surface of the device. This film allows helium to flow into the microchannels which then fill completely with the liquid due to capillary action. The liquid surface is then charged with electrons emitted from a small tungsten filament positioned several millimeters above the device. The surface electron density n_s, and so the spacing between electrons, can be controlled by the bias voltages applied to the device electrodes up to an electrohydrodynamic instability which typically occurs at densities close to 10^{10} cm^{-2} in micron-scale channels.

After the helium surface is charged, some ~10^6 electrons collect in the parallel microchannels which comprise the electron storage zone. This is confirmed by low frequency (typically 30 kHz) Sommer-Tanner ac transport measurements using dc-biased electrodes which are patterned on the base of these microchannels and are capacitively coupled to the electron system [56]. The dc bias applied to these electrodes is typically +200 mV. To allow electrons to flow from the storage zone to the trap region the bottom electrode is biased to a more positive value. Thus, the reservoir microchannel is populated with a continuous sheet of electrons with electrochemical potential equal to, or close to, the ground plane potential (0 V). The electron density in the microchannel is given to good approximation by a parallel plate capacitor model as $n_s = \varepsilon\varepsilon_0 V_b/eh$, where $\varepsilon = 1.057$ is the dielectric constant of liquid helium and ε_0 is the vacuum permittivity.

Microwave signals were transmitted to the cold stage of the cryostat through stainless steel coaxial cables and, after passing through a −20-dB cryoattenuator and a directional coupler (KRYTAR 102008020) providing a further 20-dB signal attenuation, entered the sample cell and were coupled to the resonator via the on-chip coupling port. The reflected microwave signal was amplified at room temperature with two low-noise amplifiers (Mini-Circuits ZX60-83LN-S+) giving a total +40-dB signal amplification. A vector network analyzer was used to perform the S_{11} reflection measurements with a microwave output power $P_{\mathrm{VNA}} = -35$ dBm. Each dc bias line was filtered using on-chip LC filters with cutoff frequencies close to 1 GHz.

The device was wire bonded to an FR4 printed circuit board (PCB), which was then hermetically sealed with indium wire inside the copper cell. Electrical connection to the PCB was made via mini-SMP connectors. The cell was mounted on the cold stage of the cryostat. Once at base temperature, helium was admitted to the cell from room temperature via a 1.6-mm outer diameter stainless steel tube thermally anchored at the 50, 4, and 1 K stages of the cryostat.

APPENDIX B: RESONATOR MODEL

As shown schematically in Fig. 5(a) the resonator is a split-ring CPW resonator of total length $2d = 8.72$ mm consisting of two arms, which can be viewed as two quarter-wavelength resonators capacitively coupled at the trap location and galvanically connected at the other end. The gap in the transmission line is modeled as a π network of three lumped capacitors [57] [see Fig. 5(b)]. Capacitor C_{dot} represents the coupling between excess charges at the two line ends, while C_0 models the interaction between excess strip charges and their image charges in the ground plane. In the discrete limit a lossless coplanar transmission line is modeled using distributed circuit elements as shown in Fig. 5(c), which has a unit cell containing an inductor L in series and a capacitance C to ground. For simplicity, we omit the tail CPW line shown in Fig. 1(b) and replace the connection to ground accordingly (see additional details in Ref. [34]). The Lagrangian of the transmission line in the node fluxes basis [58] is

$$\begin{aligned}\mathcal{L} = &\frac{C}{2}\sum_{n=1}^{N-1}(\dot{\phi}_n^2 + \dot{\phi}_{-n}^2)\\ &+\frac{C_0\dot{\phi}_{-N}^2}{2} + \frac{C_{\mathrm{dot}}(\dot{\phi}_N^2 - \dot{\phi}_N^2)}{2} + \frac{C_0\dot{\phi}_{-N}^2}{2}\\ &-\frac{1}{2L}\sum_{n=1}^{N-1}[(\phi_{n+1} - \phi_n)^2 + (\phi_{-n-1} - \phi_{-n})^2]. \end{aligned} \tag{B1}$$

Here, the first term describes the capacitive energy of the transmission line unit cells, the next three terms describe the contribution from the capacitances near the dot, and the final term contains all inductive energies. Using the standard approach we can define the momentum conjugate to the node fluxes as $q_n = \partial\mathcal{L}/\partial\dot{\phi}_n$, which is the charge on that node. Rewriting only the first term in Eq. (B1) in the charge basis we can write the Lagrangian of the circuit in the continuum limit as

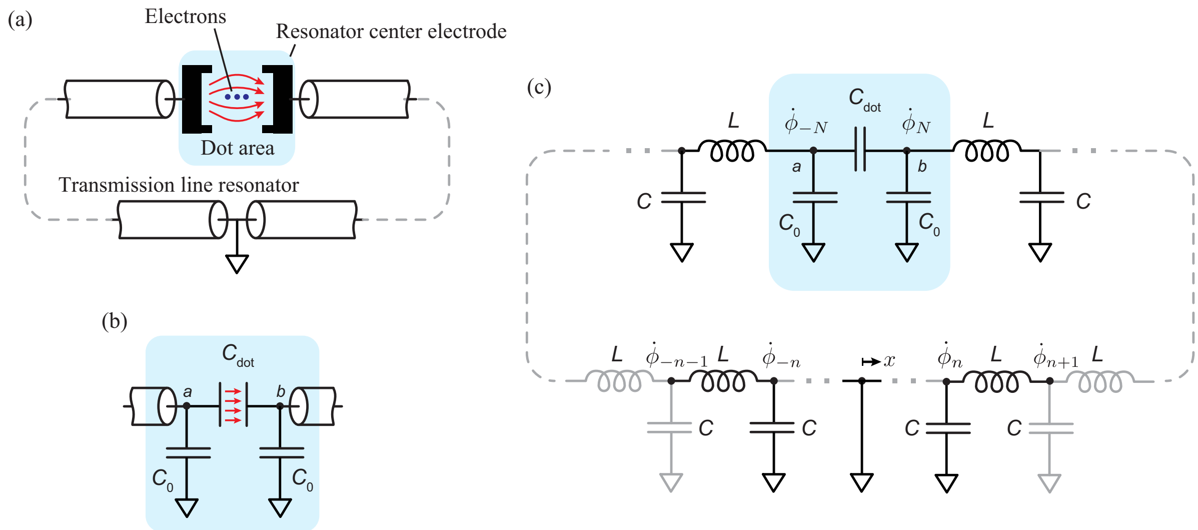

FIG. 5. (a) Schematic view of the split-ring coplanar waveguide resonator and the electron trap. (b) Capacitive π network model of the electron dot. (c) Circuit model for a split-ring transmission line resonator. Nodes are shown as dots and are labeled from $n = 0$ to $n = \pm N$. At $n = 0$ the node is galvanically connected to ground. The left and right transmission line branches are coupled via a capacitor C_{dot}. In the main text, we write the Lagrangian for this circuit model in terms of the flux node ϕ_n which is defined such that $\dot{\phi}_n$ is the voltage across the nth capacitor.

$$\begin{aligned}\mathcal{L} = {} & \frac{C_0\dot{\phi}_N^2}{2} + \frac{C_0\dot{\phi}_{-N}^2}{2} + \frac{C_{\mathrm{dot}}(\dot{\phi}_N^2 - \dot{\phi}_{-N}^2)}{2} \\ & + \int_{-d}^{0} dx \left\{ \frac{q_L^2(x)}{2C_l} - \frac{1}{2L_l}[\partial_x\phi_L(x)]^2 \right\} \\ & + \int_{0}^{d} dx \left\{ \frac{q_R^2(x)}{2C_l} - \frac{1}{2L_l}[\partial_x\phi_R(x)]^2 \right\}, \end{aligned} \tag{B2}$$

where we have used $C = dxC_l$, $L = dxL_l$ with C_l and L_l the capacitance and inductance per unit length, respectively, and $\partial_x\phi(x_n) = \lim_{dx\to 0}(\phi_{n+1} - \phi_n)/dx$. Here, negative indices are assigned to the left (L) continuum flux field and positive indices to the right (R) field. Applying Lagrange's equations of motion, the propagation along the transmission line is governed by the wave equation $v_0^2\partial_x^2\phi(x,t) - \partial_t^2\phi(x,t) = 0$, where $v_0 = 1/\sqrt{L_lC_l}$ is the speed of light in the medium. The solution of this equation can be expressed as a sum over normal modes, each oscillating at a frequency ω_m (m denotes the mode number) with a corresponding spatial wave vector $k_m = \omega_m/v_0$, which are determined by the boundary conditions. At the gapped ends of the transmission line, the current vanishes, $I(x = d) = -1/l_0\partial_x\phi(x)|_{x=d} = 0$, leading to discrete wave vectors with $k_m = \pi m/2d$ for both arms. We also note that the amplitudes at the ends of the arms are given by $u_m^L(x = -d) = \phi_{-N} = \phi_a$ and $u_m^R(x = d) = \phi_N = \phi_b$. At $x = 0$, the line is grounded, imposing a phase difference between the arms of $m\pi$. As a result, the charges in the two arms can oscillate either in phase or out of phase. These boundary conditions effectively model the two arms as a pair of $\lambda/4$ resonators.

After integration over the spatial degree of freedom in Eq. (B2) the Lagrangian can be rewritten in terms of normal modes as

$$\begin{aligned}\mathcal{L} = {} & \frac{C_0\dot{\phi}_N^2}{2} + \frac{C_0\dot{\phi}_{-N}^2}{2} + \frac{C_{\mathrm{dot}}(\dot{\phi}_N^2 - \dot{\phi}_{-N}^2)}{2} \\ & + \frac{1}{2}\sum_{m,\mu}\left[\frac{C_t\dot{\phi}_{m,\mu}^2(t)}{2} - \frac{1}{2}C_t\omega_m^2\phi_{m,\mu}^2(t)\right], \end{aligned} \tag{B3}$$

where $C_t = C_ld$ is the total capacitance of the $\lambda/4$ resonator and $\omega_m = \pi(m+1)/2d\sqrt{L_lC_l}$. Considering only the lowest mode ($m = 0$) and introducing effective transmission line inductance $L_t = 1/C_t\omega_0^2 = L_ld4/\pi^2$, the Lagrangian takes the form

$$\begin{aligned}\mathcal{L} = {} & \frac{C_0\dot{\phi}_a^2}{2} + \frac{C_0\dot{\phi}_b^2}{2} + \frac{C_{\mathrm{dot}}(\dot{\phi}_a^2 - \dot{\phi}_b^2)}{2} \\ & + \frac{1}{2}\left[\frac{C_t\dot{\phi}_a^2}{2} - \frac{\phi_a^2}{2L_t}\right] + \frac{1}{2}\left[\frac{C_t\dot{\phi}_b^2}{2} - \frac{\phi_b^2}{2L_t}\right]. \end{aligned} \tag{B4}$$

Because of large spatial nonuniformity in the system (specifically, the electron dot being much smaller than the resonator length scale, resulting in $C_0 \ll C_r$), the Lagrangian simplifies to

$$\mathcal{L} = \frac{C_{\mathrm{dot}}(\dot{\phi}_a^2 - \dot{\phi}_b^2)}{2} + \frac{C_r\dot{\phi}_a^2}{2} - \frac{\phi_a^2}{2L_r} + \frac{C_r\dot{\phi}_b^2}{2} - \frac{\phi_b^2}{2L_r}, \tag{B5}$$

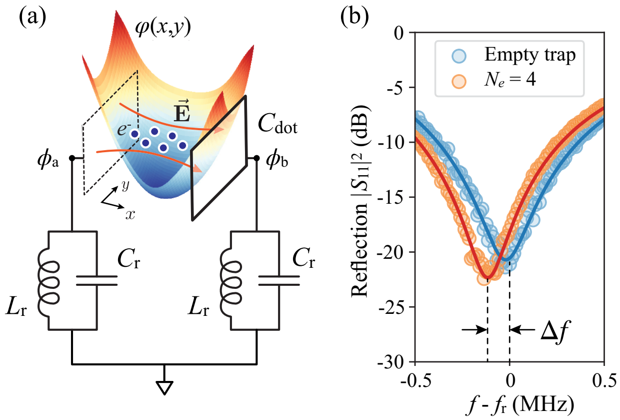

FIG. 6. (a) Schematic diagram of the circuit model used to describe the electron-resonator coupling as discussed in the main text. FEM is used to calculate the potential well which traps the electrons between the resonator arms, which are modeled as capacitor plates. (b) Experimental data showing the frequency shift Δf measured when loading the trap with $N_e = 4$ electrons. The solid lines represent fitting curves obtained using Eq. (B9).

where $C_r = C_t/2$ and $L_r = 2L_t$. This Lagrangian describes two capacitively coupled lumped LC circuits, as schematically depicted in Fig. 6(a).

Next we introduce a linear transformation to diagonalize our Lagrangian:

$$\phi_a = (\phi_+ + \phi_-)/\sqrt{2},$$
$$\phi_b = (\phi_+ - \phi_-)/\sqrt{2}, \tag{B6}$$

where ϕ_+ and ϕ_- correspond to a common and differential modes of the transmission line circuit. Under this transformation, the Lagrangian separates into a sum of two independent parts:

$$\mathcal{L} = \frac{C_r\dot{\phi}_+^2}{2} - \frac{\phi_+^2}{2L_r} + \frac{C_d\dot{\phi}_-^2}{2} - \frac{\phi_-^2}{2L_r}, \tag{B7}$$

where $C_d = C_r + 2C_{\mathrm{dot}}$. The Hamiltonian of the system can be expressed as a sum of kinetic energy written in terms of the charge variables $q_\pm = \dot{\phi}_\pm C_{r,d}$ and the potential energy expressed in terms of $\phi_\pm$ of two independent modes:

$$\mathcal{H}_{\mathrm{res}} = \frac{q_+^2}{2C_r} + \frac{\phi_+^2}{2L_r} + \frac{q_-^2}{2C_d} + \frac{\phi_-^2}{2L_r}. \tag{B8}$$

In its final form, the Hamiltonian describes two independent LC oscillators, arising from the hybridization of two $\lambda/4$ modes that are capacitively coupled at the open ends of the CPW lines. This coupling gives rise to common (+) and differential (−) modes, characterized by their ac charge eigenvectors $q_\pm = (q_a \pm q_b)/\sqrt{2}$ [34]. Here q_a and q_b are ac charges defined at the open ends [points a and b in Fig. 1(b)] of the CPW lines [see Fig. 6(a)]. For the differential mode, an ac voltage across the trap oscillates out of phase, generating an electric field $\mathbf{E}$ that excites the electron motion, thereby coupling to the electron system. The frequency of the differential mode is given by $f_r = \omega_r/2\pi = 1/(2\pi\sqrt{L_r C_d})$, where $C_d = C_l d/2 + 2C_{\mathrm{dot}}$ and $L_r = 8L_l d/\pi^2$. Here C_l and L_l are determined entirely by the geometry of the CPW lines. The width of the resonator electrode is 2 μm and the gap to the surrounding ground plane is 10 μm. Using standard methods [59] yields the values $C_d = 0.19$ pF and $L_r = 3.56$ nH with a small (×1.3) adjustment applied to L_r to better fit the measured frequencies. The cross-capacitance between the two arms of the resonator center pin electrodes $C_{\mathrm{dot}} = 20$ aF is extracted using FEM calculations and is primarily governed by the geometry of the trap. The common mode frequency $f_c = \omega_c/2\pi = 1/2\pi\sqrt{1/(L_r + 2L_{\mathrm{tail}})C_r} < 4$ GHz is suppressed by engineering a large tail inductance L_{tail} using a long CPW line of length ~2.3 mm connected at point c [34]. To bias the resonator, the voltage V_r is applied at the opposite end of the tail CPW line.

The resonator is measured in reflection, with capacitive coupling to the drive port through a small section of one of the resonator arms, as illustrated in Fig. 1(b). The normalized reflection coefficient S_{11} near the resonance can be expressed as [60]

$$S_{11} = 1 - \frac{2Q_t}{Q_c}\frac{ae^{i\theta}}{1 + i2Q_t\delta}, \tag{B9}$$

where $Q_t = (1/Q_i + 1/Q_c)^{-1}$ is the total quality factor, Q_i and Q_c are the internal and coupling quality factors, respectively, $\delta = (f - f_r)/f_r$, and the complex quantity $ae^{i\theta}$ accounts for asymmetries of the reflection from the resonator due to impedance mismatches in the measurement circuit or interference from background signals caused by leakage between ports in the directional coupler. We note that the intrinsic quality factor of the resonator accounts for both cavity decay and dephasing. This is valid under the weak or dispersive coupling approximation, where two-level fluctuators (located in the dielectric substrate or thin oxide films on the resonator electrodes) induce random resonance frequency shifts through stochastic switching [61]. A typical measured reflection spectrum displays a dip in magnitude near the resonance frequency as shown in Fig. 2(b). Fitting to this data gives the resonance frequency of the differential mode $f_r = 6.04383$ GHz with internal quality factor $Q_i = 5800$ and coupling quality factor $Q_c = 4800$. These measurements were performed at $T = 1.1$ K before charging the helium surface. We note that the experiments shown here were successfully repeated with several devices, each with different but quantitatively similar values of f_r and Q_i.

APPENDIX C: ELECTRON-RESONATOR COUPLING MODEL

In our experiments, electrons are loaded into an electrostatic trap created at the open ends of the resonator electrodes. The trap potential $\varphi(x, y)$ is engineered to facilitate motion along the direction of the resonator electric field, in this case the x axis. The trap potential $\varphi(x, y) = \sum_k \alpha_k(x, y) V_k$ is determined by dimensionless functions α_k representing the potential due to electrode k at unit voltage, which we calculate from FEM. For the temperatures relevant to our experiments, where $k_B T > \hbar\omega_e$ (with ω_e representing the characteristic single-electron motion frequency determined by trap potential curvature), we describe the nondegenerate electron system using classical equations of motion in a long-wave approximation. The full Hamiltonian of the system,

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_{\text{el}} + \mathcal{H}_{\text{res}} + \mathcal{H}_{\text{el-res}}, \tag{C1}$$

where $\mathcal{H}_{\text{el}}$ is the Hamiltonian of the electron subsystem, $\mathcal{H}_{\text{res}}$ is the Hamiltonian of the resonator subsystem [see Eq. (B8)], and $\mathcal{H}_{\text{el-res}}$ is the electron-resonator coupling term. We assume that the anisotropy of the trapping potential allows the system to be treated as one dimensional along the x axis; therefore, we write the Hamiltonian for in-plane electron motion:

$$\mathcal{H}_{\text{el}} = \frac{1}{2m_e}\sum_i p_i^2 - e\sum_i \varphi(x_i) + \frac{1}{2}\frac{e^2}{4\pi\epsilon_0}\sum_i\sum_{j\neq i}\frac{1}{|x_i - x_j|}.$$

p_i is the momentum of electron i in the x direction, and last term describes Coulomb interaction between electrons. Expanding this Hamiltonian to second order around the equilibrium positions x_i results in

$$\mathcal{H}_{\text{el}} = \frac{1}{2m_e}\sum_i p_i^2 + \frac{e}{2}\sum_i \frac{\partial^2\varphi}{\partial x^2}\bigg|_{x_i}\delta x_i^2 + \frac{m_e}{2}\sum_i\sum_{j\neq i}\kappa_{ij}(\delta x_i - \delta x_j)^2, \tag{C2}$$

where δx_i is the electron displacement and $\kappa_{ij} = e^2/4\pi\epsilon_0 m_e r_{ij}^3$ characterizes the Coulomb interaction strength with $r_{ij} = |x_i - x_j|$ being the pairwise distance. Here, energy offset terms are omitted as they do not affect electron dynamics, and the linear terms cancel due to the force balance condition at equilibrium. We note that the coupling term κ_{ij} represents unscreened Coulomb interactions, which can be reduced in our trap geometry due to screening effects from nearby electrodes.

The interaction energy between an electron and resonator photons arises from the electron's potential energy in the electric field generated by voltages at the ends of the resonator electrodes $\dot{\phi}_a$ and $\dot{\phi}_b$, and is given by

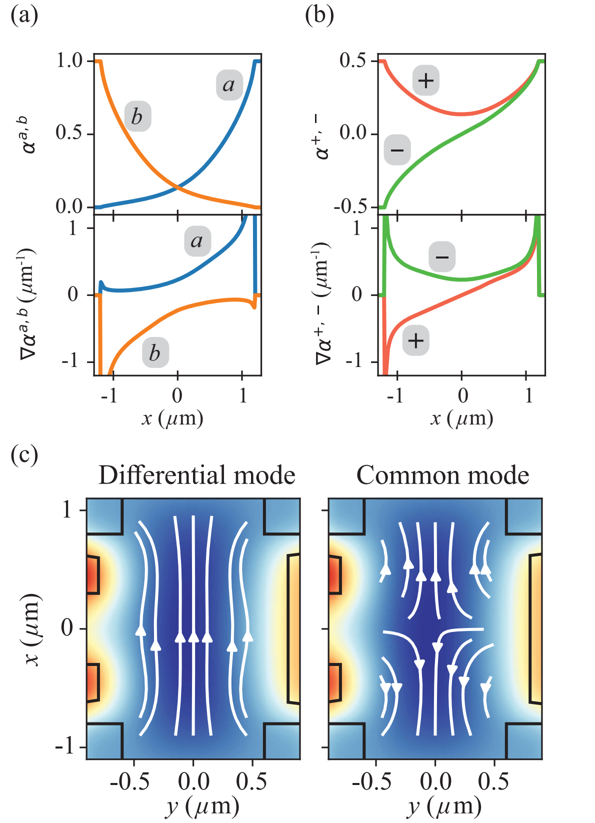

FIG. 7. (a) Coupling constant distributions $\alpha^{a,b}(x)$ and their corresponding derivatives $\nabla\alpha^{a,b}(x)$ for the resonator electrodes calculated using FEM simulations. (b) Coupling constants and their derivatives associated with the differential (−) and common (+) modes of the resonator. (c) Electric field lines (white arrows) in the trap region created by the differential (left) and common (right) modes of the resonator.

$$\mathcal{U}_{\text{el-res}} = -e\sum_i(\alpha^a(x_i)\dot{\phi}_a + \alpha^b(x_i)\dot{\phi}_b), \tag{C3}$$

where $\alpha^{a,b}$ are dimensionless coupling constants determined via FEM simulations by setting the respective resonator electrode (a or b) to 1 V while grounding all other electrodes [see Fig. 7(a)]. Expanding the potential energy around the equilibrium position of electrons and omitting the constant term gives

$$\mathcal{U}_{\text{el-res}} = -e\sum_i(\nabla\alpha_i^a\dot{\phi}_a + \nabla\alpha_i^b\dot{\phi}_b)\delta x_i, \tag{C4}$$

where we have used the notation $\nabla\alpha_i = \partial\alpha/\partial x|_{x_i}$. Next, rewriting the interaction energy in terms of the resonator normal modes using Eq. (B6), and defining $\alpha^{\pm} = (\alpha^a \pm \alpha^b)/2$ [see Fig. 7(b)], we obtain the final form of the interaction Hamiltonian in the charge basis of the resonator:

$$\mathcal{H}_{\text{el-res}} = -\sqrt{2}e\sum_i \left(\frac{\nabla\alpha_i^+ q_+}{C_r} + \frac{\nabla\alpha_i^- q_-}{C_d}\right)\delta x_i, \tag{C5}$$

In this expression, the electrons couple to both the differential and common modes of the resonator. The resonator was designed such that the common mode is shifted to lower frequencies, outside the measurement band. In the experiments, we primarily drive the system at frequencies near the differential mode of the resonator. Therefore in the following we focus solely on the differential mode of the resonator and replace δx_i with x_i for simplicity.

To proceed with the derivation of the equations of motion for the full coupled system, we write the Hamilton equations of motion for both the resonator and electron:

$$\dot{\phi}_- = \frac{\partial\mathcal{H}_{\text{tot}}}{\partial q_-}, \qquad \dot{p}_i = \frac{\partial\mathcal{H}_{\text{tot}}}{\partial x_i},$$
$$\dot{q}_- = -\frac{\partial\mathcal{H}_{\text{tot}}}{\partial\phi_-}, \qquad \dot{x}_i = -\frac{\partial\mathcal{H}_{\text{tot}}}{\partial p_i}. \tag{C6}$$

We now present the final form of the coupled equations of motion (EOM) for the resonator and the electrons:

$$\ddot{q}_- + \omega_r^2 q_- - \sqrt{2}e\omega_r^2\sum_i \nabla\alpha_i^- x_i = 0,$$
$$\ddot{x}_i + \omega_i^2 x - \sum_j \kappa_{ij}x_j - \frac{\sqrt{2}e\nabla\alpha_i^-}{m_e C_d}q_- = 0, \tag{C7}$$

where we have defined the resonator frequency $\omega_r^2 = 1/L_r C_d$, and the frequency of motion for the ith electron near its equilibrium position is given by $\omega_i^2 = e/m_e\ \partial^2\varphi/\partial x^2 + m_e\sum_j \kappa_{ij}$. EOM for the electron subsystem can be written in a matrix form:

$$\ddot{\vec{\mathbf{x}}} + \mathbb{M}\vec{\mathbf{x}} = \frac{\sqrt{2}e}{m_e C_d}\vec{\mathcal{E}}q_-,$$
$$\vec{\mathbf{x}} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix}, \qquad \vec{\mathcal{E}} = \begin{pmatrix} \nabla\alpha_1^- \\ \nabla\alpha_2^- \\ \vdots \\ \nabla\alpha_N^- \end{pmatrix},$$
$$\mathbb{M} = \begin{pmatrix} \omega_1^2 & -\kappa_{12} & \cdots & -\kappa_{1N} \\ -\kappa_{21} & \omega_2^2 & \cdots & -\kappa_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ -\kappa_{N1} & -\kappa_{N2} & \cdots & \omega_N^2 \end{pmatrix}. \tag{C8}$$

The EOM for the coupled resonator-electron system can now be expressed in a compact form:

$$\ddot{q}_- + \omega_r^2 q_- - \sqrt{2}e\omega_r^2(\vec{\mathcal{E}}\cdot\vec{\mathbf{x}}) = \frac{V_d}{L_r}e^{i\omega t},$$
$$\ddot{\vec{\mathbf{x}}} + \mathbb{M}\vec{\mathbf{x}} = \frac{\sqrt{2}e}{m_e C_d}\vec{\mathcal{E}}q_-, \tag{C9}$$

where we have included the resonator drive term $V_d e^{i\omega t}$. We assume a solution of the form $[q(t), \vec{\mathbf{x}}(t)] = [A_q, \mathbf{A}_x]e^{i\omega t}$. Substituting this ansatz into the EOM and applying a normal (*vibrational*) mode decomposition to the electronic subsystem—where the vibrational modes satisfy the eigenvalue equation $\mathbb{M}\vec{\mathbf{x}}_n = \omega_n^2\vec{\mathbf{x}}_n$, with ω_n and $\vec{\mathbf{x}}_n$ representing the eigenfrequency and normalized eigenvector of the nth mode, respectively—yields the electron system response function in the form

$$\mathbf{A}_x = q_-\frac{\sqrt{2}e}{m_e C_d}\sum_n \frac{(\vec{\mathcal{E}}\cdot\vec{\mathbf{x}}_n)}{\omega_n^2 - \omega^2 + 2i\omega\Gamma_n}\vec{\mathbf{x}}_n, \tag{C10}$$

where the dot product $(\vec{\mathcal{E}}\cdot\vec{\mathbf{x}}_n) = \sum_i \nabla\alpha_i^- x_{i,n}$ can be interpreted as an effective electric field strength (defined per unit voltage) that drives the nth vibrational mode. Here, $x_{i,n}$ denotes the normalized amplitude of the electron displacement in the nth mode. We also include a phenomenological damping constant Γ_n associated with the nth vibrational mode, which describes the energy relaxation due to interactions between the electron system and ^{4}He vapor atoms, quantized He surface capillary waves (ripplons), and bulk phonons [20]. The resonator response function is obtained by substituting Eq. (C10) into Eq. (C9) and incorporating intrinsic losses in the resonator as a damping term κ, resulting in the following form:

$$A_q = \frac{V_d}{L_r}\frac{1}{\omega_r^2 - \omega^2 + 2i\omega\kappa - \omega_r^2\chi_e(\omega)}, \tag{C11}$$

where we have defined the linear electric susceptibility (or ac current susceptibility) of the electron system,

$$\chi_e(\omega) = \sum_n \frac{4g_n^2}{\omega_n^2 - \omega^2 + 2i\omega\Gamma_n}, \tag{C12}$$

where g_n represents the coupling strength between the microwave field and the nth vibrational mode of the electron system, and ω_n is the eigenfrequency of that mode. The coupling strength g_n determines the dipole energy of the vibrational mode and is defined by the dot product of the field strength vector $\vec{\mathcal{E}}$ and the normalized eigenvector $\vec{\mathbf{x}}_n$ of the specific vibrational mode:

$$g_n = \sqrt{2}\frac{e(\vec{\mathcal{E}}\cdot\vec{\mathbf{x}}_n)}{2\sqrt{m_e C_d}}. \tag{C13}$$

With these definitions of g_n and $\vec{\mathcal{E}}$, Eq. (C12) generalizes to the 2D and even 3D bound case. We note that in a uniform electric field and without interactions, the coupling strength is given by $g_n = \sqrt{N} g_{\mathrm{e}}$, where g_{e} represents the coupling energy for a single electron, which exhibits the $\sqrt{N}$ scaling.

Driving the system near the resonance $\omega \approx w_r$ allows the resonator response to be simplified further:

$$A_q \approx \frac{V_d}{2Z_r} \frac{1}{\omega_r - \omega + i\kappa - \omega_r \chi_e(\omega)/2}. \tag{C14}$$

In the weak coupling limit, where the shift of the response peak from the bare resonator frequency is small (i.e., $\Delta\omega/\omega_r \ll 1$ or $\chi_e \ll 1$), the response function of the resonator exhibits a Lorentzian profile, with its resonance frequency slightly displaced from the bare value ω_r by an amount $2\pi\Delta f \approx -\omega_r \mathrm{Re}\{\chi_e(\omega_r)/2\}$.

The expression for the coupling also indicates which vibrational modes can be excited by a particular mode of the resonator, as the field vector depends on the specific resonator mode being driven. This relationship can be readily deduced from the symmetries of the vibrational mode of the electron system and the resonator modes. Electric field distributions of the resonator modes are shown in Fig. 7. For the differential mode, the generated electric field is symmetric about the x axis and characterized by the distribution $\nabla\alpha^-$. As a result, it couples more effectively to vibrational modes of the electron system whose displacement vectors are predominantly aligned in a single direction, i.e. those with a nonzero center of mass displacement. In contrast, the electric field associated with the common mode is symmetric about the origin, making it more likely to couple to electron motion characterized by out-of-phase oscillations in the $x > 0$ and $x < 0$ regions of the trap. Equation (C12) clearly shows that the largest response in the frequency shift will be induced by vibrational modes where the eigenfrequencies are closest to the resonator frequency and the dipole coupling g_n is large.

The response of a single electron in a symmetrical trap can be readily derived using Eqs. (C12) and (C13). Single-electron motion in the trap is solely determined by the electrostatic trap curvature $\omega_e^2 = e m_e^{-1} [d^2\varphi(x,y)/dx^2]_{x,y=0}$, which is controlled by gate voltages. The coupling strength simplifies to $g_{\mathrm{e}} = \sqrt{2} e \mathcal{E}_x / 2\sqrt{m_e C_d}$, which coincides with the expression for the dipole energy $e\mathcal{E}_x \hat{x} \hat{V}$ of zero-point fluctuations of the electron motion $\hat{x}$ and vacuum fluctuations of the resonator field $\mathcal{E}_x \hat{V}$. Here $\mathcal{E}_x = \nabla\alpha^-$ is the electric field strength per unit voltage created by the differential mode of the resonator at $x = 0$. We note that driving electron motion with the differential mode gives $\sqrt{2}$ enhancement in the coupling compared to a single-mode resonator. For the trap design shown in Fig. 1(c) we find $\mathcal{E}_x = 0.23\ \mu\mathrm{m}^{-1}$ which gives single electron-cavity photon coupling energy $g_{\mathrm{e}}/2\pi = 9.8$ MHz. It is evident that for the common mode $\nabla\alpha^+ = 0$ at the center of the trap due to its symmetry [see Fig. 7(b)]. Consequently, this resonator mode cannot excite single electron motion. For larger numbers of electrons in the arbitrary trap potential $\varphi(x, y)$ we determine the eigenmodes and equilibrium configuration of interacting electrons by numerical minimization procedures [34,35]. Figure 6(b) shows measured spectra both without electrons and with $N_e = 4$ electrons in the trap. The observed frequency shift, $\Delta f \approx -100$ kHz, agrees well with the numerically calculated value of $\Delta\omega/2\pi = -95$ kHz. The parameters used to calculate the frequency shifts are summarized in Table I, with the corresponding voltages provided in the captions of Figs. 2, 3, and 4 of the main text.

TABLE I. List of parameters (including offsets on the voltages controlling the trap electrostatic potential) used to estimate the susceptibility of electron system and the frequency shift of the microwave resonator.

		Value	Offsets
f_r	Measured	6.04383 GHz	⋯
$\Gamma_n/2\pi$	Fit	1–2 GHz	⋯
C_d	Geometric	0.19 pF	⋯
V_{b}	Applied	⋯	⋯
V_{r}	Applied	⋯	[−50, −80] mV
V_{sg}^u	Applied	⋯	+10 mV
V_{sg}^l	Applied	⋯	−10 mV
V_{un}	Applied	⋯	⋯

APPENDIX D: TWO ELECTRON MOTIONAL MODES

Here we present the derivation of analytical expressions for a system of two electrons coupled to a resonator field. We start with the Hamiltonian describing a 1D motion of two electrons within a trapping potential:

$$\mathcal{H}_e = \sum_{i=1,2} \left[\frac{p_i^2}{2m_e} - e\varphi(x_i) \right] + \frac{e^2}{4\pi\epsilon_0} \frac{1}{|x_1 - x_2|}. \tag{D1}$$

The equilibrium positions of the electrons x_0 are determined from the force balance equation $\partial\varphi/\partial x|_{x=x_0} = e/(4\pi\epsilon_0 d_e^2)$, where $d_e = 2x_0$ is the distance between the electrons. Subsequently, we perform a Taylor expansion of both the potential energy and the Coulomb energy in terms of the displacements δx_i around the equilibrium positions, which results in the Hamiltonian:

$$\mathcal{H}_e = \sum_{i=1,2} \left(\frac{p_i^2}{2m_e} + \frac{m_e \omega_e^2}{2} \delta x_i^2 \right) + \frac{m_e \omega_C^2}{2} (\delta x_1 - \delta x_2)^2, \tag{D2}$$

where $\omega_C^2 = 2e^2/4\pi\epsilon_0 m_e d_e^3$ and we have used the relation between the curvature of the potential and the single-electron frequency $\partial^2\varphi/\partial x^2|_{x=\pm x_0} = -m_e\omega_e^2/e$ valid in the harmonic approximation of the trap potential. We have also omitted constant terms, and the linear terms in displacement cancel out due to the force balance condition. This Hamiltonian can be diagonalized using the following transformation:

$$\begin{aligned} x_+ &= (\delta x_1 + \delta x_2)/\sqrt{2}, \\ x_- &= (\delta x_1 - \delta x_2)/\sqrt{2}. \end{aligned} \tag{D3}$$

The new coordinates correspond to two vibrational modes, which are the in-phase (+) and out-of-phase (−) motion with eigenvectors $\vec{\mathbf{x}}_\pm = \binom{1}{\pm 1}/\sqrt{2}$. As $x_+^2 + x_-^2 = \delta x_1^2 + \delta x_1^2$, the two electron Hamiltonian takes the final form:

$$\mathcal{H}_e = \frac{p_+^2}{2m_e} + \frac{m_e\omega_{2+}^2 x_+^2}{2} + \frac{p_-^2}{2m_e} + \frac{m_e\omega_{2-}^2 x_-^2}{2}, \tag{D4}$$

where the frequency of the in-phase and out-of-phase motion is given by $\omega_{2+}^2 = \omega_e^2$ and $\omega_{2-}^2 = \omega_e^2 + 2\omega_C^2$, respectively.

The in-phase electron motion couples to the differential mode, which is evident from the nonzero value of the dot product $(\vec{\mathcal{E}}\cdot\vec{\mathbf{x}}_+)$ with electric field strength vector defined as $\vec{\mathcal{E}} = \sqrt{2}\mathcal{E}_2\binom{1}{1}$. In this case both electrons experience an equal field strength $\mathcal{E}_2 = [(\nabla\alpha^a - \nabla\alpha^b)/2]_{x=r_{12}/2}$ aligned with their displacement vectors. In the harmonic approximation this results in a two electron-resonator coupling energy $g_{2+} = \sqrt{2}g_e$. This leads to values of Δf that are twice larger than the single-electron case. From the dot product $(\vec{\mathcal{E}}\cdot\vec{\mathbf{x}}_-)$ it is clear that the out-of-phase motion of two electrons can only be excited by the common mode of the resonator.

APPENDIX E: METASTABLE STATES

The correlated electron system confined within the trap may potentially exhibit metastable configurations. If the thermal energy is sufficient to induce transitions between the ground and these metastable states, then the system response would be described by a statistical average of the responses from each of these metastable and ground-state configurations.

To address this issue, we performed numerical simulations to explore the possible electron configurations within the electrostatic trapping potential. To identify the ground-state configuration, we started with randomly initialized electron positions within the trap and employed gradient descent methods to find the minimum of the potential energy of the system. Additional annealing steps were included to minimize the likelihood of the algorithm becoming trapped in local minima. To explore metastable states, the annealing steps were disabled and the resulting configurations were analyzed. This procedure was repeated 2000 times to gather statistical data on the metastable states. The results are summarized in Fig. 8. Figure 8(a) shows the energy spectrum of the electron system $\Delta E(N_e)$ offset by the corresponding ground-state energy of each electron number N_e in the trap. For voltage configurations used in Fig. 3 of the main text, the resulting potential is typically anisotropic, exhibiting weaker confinement along the x axis. For a small number of electrons N_e, this results in a single-chain configuration of electrons aligned along

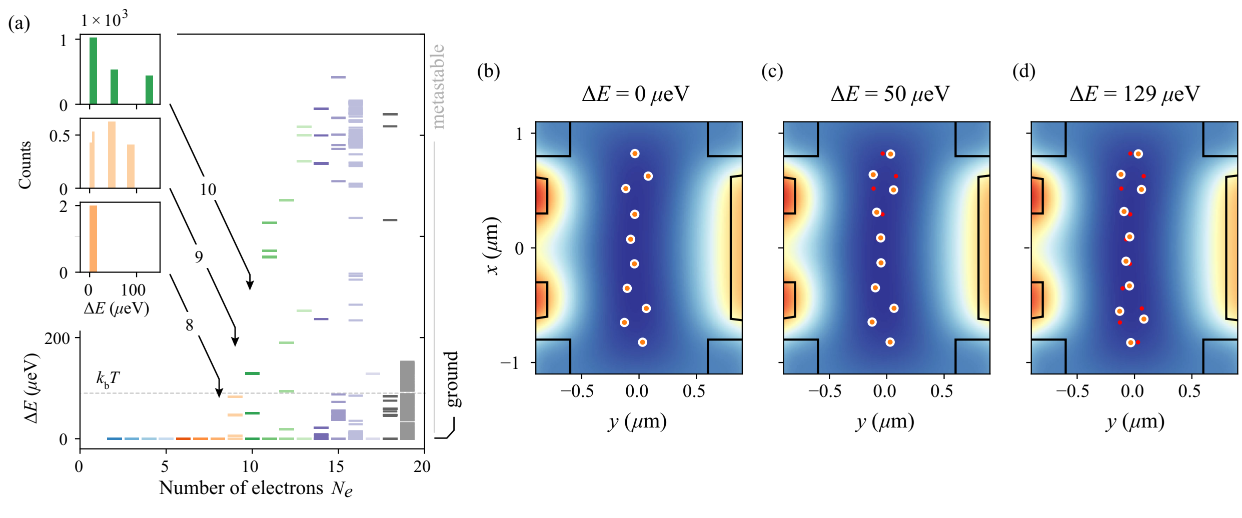

FIG. 8. (a) The energy spectrum of the electron system $\Delta E(N_e)$ for varying numbers of electrons in the trap, where metastable states correspond to configurations with $\Delta E > 0$. The inset displays a histogram of likelihood of the system settling into different local and global energy minima during the search for $N_e = 8, 9, 10$. The horizontal dashed line indicates the thermal energy of electrons under experimental conditions. Panels (c) and (d) show examples of metastable configurations, while panel (b) presents the ground-state configuration for $N_e = 9$.

the x axis. In this regime, all numerical simulations converge to the ground-state configuration, and no metastable states are observed [see inset in Fig. 8(a)]. As the number of electrons increases, Coulomb repulsion drives a transition to zigzaglike configurations, and low-energy metastable states begin to appear in the spectrum [see an example of metastable states for $N_e = 9$ in Figs. 8(b)–8(d)]. The increased likelihood of the system occupying these metastable states is reflected in the histogram of their occurrence shown in the inset of Fig. 8(a) for $N_e = 8, 9, 10$. As the number of electrons increases further, the number of accessible metastable states also grows. However, the main data presented in the text focus on systems with a small number of electrons $N_e \leq 4$, for which metastable states are absent given the specific trap geometry and potential configurations used in the experiments. We also employed an alternative approach based on simulating the damped equations of motion. Both methods yielded consistent results.

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